G H Raisoni International Skill Tech University, Pune

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End Semester Examinations - Winter Term 2025

School of Engineering and Technology

F.Y. B.Tech., Semester - I (All)

Course Name: Environmental Chemistry | **Course Code: 25UBSL1102**

Max Marks: 60 | **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D is compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Discuss the process of water softening by zeolite method with neat labelled diagram and chemical reaction. Mention two advantages of this process.	5	1	1
	B	If 40 ml of water sample requires 16 ml of 0.04 M EDTA during titration. Whereas 50 ml of boiled water sample requires 10.5 ml of same EDTA in the titration. Determine total, alkaline and non-alkaline hardness of the water sample. 1600 160 840	5	1	3
	C	Explain the different steps involved in the purification of domestic water.	5	1	2
	D	Define temporary and permanent hardness of water.	2	1	1
2		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Explain the various steps involved in the solid waste management in detail.	5	2	2
	B	Describe biomedical waste, providing suitable examples and associated hazards.	5	2	1
	C	Explain any two major sources of solid waste in detail.	5	2	1
	D	Explain the harmful effects of electronic solid waste.	2	2	1
3		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			

	A	Describe the construction & working of Lead acid Battery along with diagram & chemical reactions involved at anode and cathode.	5	3	
	B	Define a battery and distinguish between primary and secondary batteries.	5	3	1
	C	Describe the construction & working of Daniel cell along with diagram & chemical reactions involved at anode and cathode.	5	3	2
	D	Define reversible and irreversible batteries.	2	3	1
4		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Define fuel and distinguish between conventional and non-conventional energy sources with example.	5	3	1
	B	Observations in the Boy's Calorimeter experiment on gaseous fuel are given below, find GCV and NCV of the fuel. Volume of gas burnt (STP) $0.08m^3$ Mass of cooling water used=29.5 kg Rise in temperature of circulating water=$9.1\ ^0$ C Mass of condensed steam =0.04kg 30 62.125	5	3	3
	C	Calculate the GCV (in kJ/kg) 0.80 gm of a fuel containing 80% of carbon, when burnt in a bomb calorimeter increased the temperature of water from 28.5 to 30.7^0 C. The calorimeter contains 230 grams of water and it's water equivalent is 200 grams.	5	3	3
	D	Write down the any four characteristics of good fuel.	2	3	1
5		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Define air pollution and describe any four methods used to control air pollution.	5	4	2
	B	Define Water pollution and describe any two major sources of water pollution in brief.	5	4	2
	C	Describe the three main sources of air pollution and explain their effects.	5	4	2
	D	Enlist any four major air pollutants.	2	4	1

- End of the Question Paper -